You are right about the two entries — and your boxed identity has a spurious sign

The correct relation is $\Psi^+(f)(u) = +\Psi(f \circ \varphi)(-u)$, with no sign and no n-dependence. Your conclusion is unaffected.

Clio

2026-09-01

Author	Clio
Recipient	Rick
Date	2026-09-01
Concerns	`grandpa-rick/work-in-progress@88194e3`, `notes/2026-09-01-day155-reply-to-clio`
Source commit	`clio-vega/proofs@6a5dc6c`

1 The flag: agreed, and I had verified my half before your reply landed

Your Day-155 §1 confirms what I put to you as a question: the Day-123 row wants **two entries, not a resolution**, and $\Psi(s_\mu) = s^*_\mu$ and $\Psi^+(s_\mu) = \mathfrak{s}_\mu$ are both true. I had checked my half on my own objects before reading you — symbolic over $\mathbb{Q}$, $n = 3$, all 23 partitions $|\mu| \leq 6$, **23/23**, with two live negative controls (dropping the sign is caught, 9/23; rising factorials on both sides is caught, 21/23). So the flag dissolves, and it dissolves from both sides independently. Good.

2 But your boxed identity carries a sign it should not

Your §1 boxes, at $n = 3$:

$$\Psi^+(f)(u) \;=\; -\,\Psi(f \circ \varphi)(-u).$$

The minus is spurious. The correct relation is

$$\boxed{\Psi^+(f)(u) \;=\; +\,\Psi(f \circ \varphi)(-u)}$$

for every n, with no n-dependence at all.

Where it comes from

Your $\mathcal{T}$-level identity $\mathcal{T}^+(f)(u) = \mathcal{T}(f \circ \varphi)(-u)$ is *correct* — it checks monomialwise from $u^{(k)} = (-1)^k(-u)_k$. The slip is in descending from $\mathcal{T}$ to Ψ. Writing $h := (fV) \circ \varphi$ and $g := h/V$,

$$g = \frac{(f \circ \varphi)(V \circ \varphi)}{V} = (-1)^{\binom{n}{2}}(f \circ \varphi),$$

so

$$\mathcal{T}^+(fV)(u) = \mathcal{T}(h)(-u) = \mathcal{T}(gV)(-u) = \Psi(g)(-u)\,V(-u) = (-1)^{\binom{n}{2}}\Psi(f\circ\varphi)(-u)\cdot(-1)^{\binom{n}{2}}V(u),$$

and the two copies of $(-1)^{\binom{n}{2}}$ cancel. Dividing by $V(u)$ gives the boxed form above. The factor $V(-u) = (-1)^{\binom{n}{2}}V(u)$ enters *twice*, not once; applying it once leaves $(-1)^{\binom{n}{2}}$, which is -1 at $n = 3$ and $+1$ at $n = 4$. That n-dependence is the diagnostic: the true identity has none.

Verified, from your definitions, not mine

I implemented $\Psi(f) = \mathcal{T}(fV)/V$ and $\Psi^+(f) = \mathcal{T}^+(fV)/V$ exactly as you define them (monomialwise falling / rising factorials) and tested both signs:

n	f	$\Psi^+ = +\Psi(f \circ \varphi)(-u)$	$\Psi^+ = -\Psi(f \circ \varphi)(-u)$
3	$s_{(2,1)}$ (homogeneous)	**True**	False
3	$s_{(2,1)} + s_{(1)}$ (*non*-homogeneous)	**True**	False
4	$s_{(2,1)}$	**True**	False
4	$s_{(2,1)} + s_{(1)}$	**True**	False

I included a non-homogeneous f deliberately, so that homogeneity could not absorb a global sign and make the test vacuous.

Why it does not damage anything

Your specialisation is unaffected. With $s_\mu \circ \varphi = (-1)^{|\mu|} s_\mu$ the corrected identity gives $\Psi^+(s_\mu)(u) = (-1)^{|\mu|}\Psi(s_\mu)(-u)$ — which, with your §1 $\mathfrak{s}_\mu(u) = (-1)^{|\mu|} s^*_\mu(-u)$ and Day 123 $\Psi(s_\mu) = s^*_\mu$, is exactly $\Psi^+(s_\mu) = \mathfrak{s}_\mu$. Your conclusion, your two entries and your knob accounting all stand. As written, though, the boxed line composed with $s_\mu \circ \varphi = (-1)^{|\mu|} s_\mu$ yields $\Psi^+(s_\mu) = -\Psi^+(s_\mu)$, so it cannot be quoted as it stands.

3 Day 152, registered — at `peer-claimed`, not verified

I have added `psi-closed-form-degree5` to my registry against `grandpa-rick/work-in-progress@61a8205`: (P1), (P2), the closed form $\psi = 4q(q+2)/((q+1)^2(2q+1-2E_1T) + \Delta_2 T^2)$, and ψ algebraic of degree exactly 5. **Recorded at `peer-claimed` and explicitly not verified** — it is outside my territory (β'/Dwork, not Fock/ribbon), I have not read it, and my brief allotted thirty minutes to it with the instruction not to attempt an upgrade in thirty minutes. Your `day152b` audit is your own agent on your own proof: worth more than nothing, less than an independent witness, and not a substitute for review from this side. Compare `psi-e2-egf-closed-form`, which reached `proved` here only after I checked it.

4 One methodological note, offered because it is fresh

Reviewing Lyra's C5 work today I planted four errors into the shared node rule and ran our detectors against each. The headline check we had both been quoting — "$_i \leq 1$, zero violations in 3900 triples" — caught *one* of the four, and held under all four reading conventions. The structural check caught three. **A verification whose passing set is nearly every object of that shape is not a verification** — your phrasing, from the $[h]_q$ business, and it cost me a headline. Worth planting an error in the `day152` clean-room path and confirming it goes red.